



Diffusion-driven Incomplete Multimodal Learning for Air Quality Prediction

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Predicting air quality using multimodal data is crucial to comprehensively capture the diverse factors influencing atmospheric conditions. Therefore, this study introduces a multimodal learning framework that integrates outdoor images with traditional ground-based observations to improve the accuracy and reliability of air quality predictions. However, aligning and fusing these heterogeneous data sources poses a formidable challenge, further exacerbated by pervasive data incompleteness issues in practice. In this article, we propose a novel incomplete multimodal learning approach (iMMAir) to recovery missing data for robust air quality prediction. Specifically, we first design a shallow feature extractor to capture modal-specific features within the latent representation space. Then we develop a conditional diffusion-driven recovery module to mitigate the distribution gap between the recovered and true data. This module further incorporates two conditional constraints of temporal correlation and semantic consistency for effective modal completion. Finally, we reconstruct incomplete modalities and fuse available data using a multimodal transformer network to predict the air quality. To alleviate the modality imbalance problem, we employ an adaptive gradient modulation strategy to adjust the optimization of each modality. Experimental results demonstrate that iMMAir significantly reduces prediction errors, outperforming baseline models by an average of 5.6% and 2.5% in air quality regression and classification tasks. Our source code and data are available at <https://github.com/pestasu/iMMAir>.

CCS Concepts: • **Computer systems organization** → **Sensor networks**; • **Computing methodologies** → **Artificial intelligence**;

Additional Key Words and Phrases: Incomplete multimodal learning, air quality prediction, diffusion model

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1 Introduction

Air quality prediction is crucial for efficient urban management and safeguarding human well-being [12, 37, 43, 48]. However, estimating atmospheric conditions in urban areas involves a complex interaction of multiple factors, such as vehicular emissions, industrial activity, and changing weather patterns. To effectively address these complexities, it is imperative to integrate diverse data sources for accurate and prompt air quality predictions [16, 22, 23, 36].

Traditional air quality prediction methods primarily rely on ground-based observations collected from static monitoring sites [13–15, 27], which often proves inadequate in capturing the complex array of environmental variables that affect air quality. Recent research has demonstrated the feasibility of leveraging outdoor images captured by mobile devices to estimate air quality [4, 28, 29], which captures crucial optical information related to air pollution levels, such as visibility and particulate matter. Furthermore, Rowley and Karakuş [42] integrate satellite images to explore pollutant distribution across extensive geographic areas. In contrast, leveraging outdoor images offers a more accessible and user-centric alternative for general public. Therefore, we fuse outdoor images with traditional ground-based observations into a multimodal framework to enhance prediction performance while maintaining cost-effectiveness.

However, the alignment and integration of these heterogeneous data sources often encounter challenges of modal incompleteness due to device deployment and sensor malfunctions, as illustrated in Figure 1(a). For instance, image data is completely absent near the monitoring site, or sequence data is partially missing. Recent recovery-based methods for incomplete multimodal learning in computer vision [5, 53, 60] generally employ deep neural networks to estimate missing modalities based on the available ones. Nevertheless, these approaches often face limitations in scenarios missing partial data within a modality and struggle to ensure the fidelity of recovered data. In addition, each data modality exhibits distinct characteristics. Specifically, air quality data quantifies pollution concentrations, weather data reflects atmospheric conditions, and outdoor images capture optical variations. Existing multimodal approaches have primarily focused on incorporating diverse sensory data streams as input channels or integrating feature representations from various encoding strategies [29, 59]. While these methods effectively capture spatiotemporal correlations within individual modalities, they often neglect the interactions between different modalities. Furthermore, simply incorporating diverse data modalities into a model does not necessarily guarantee the inherent predictive abilities of each modality. Peng et al. [38] reveal that even though multimodal models may outperform their unimodal ones, they often fail to exploit the potential of each modality. Consequently, merely treating multisource as homogeneous inputs fails to adequately explore their potential for complementarity in multimodal air quality prediction, as shown in Figure 1(b). Therefore, there is a pressing need to develop an incomplete multimodal learning approach specifically tailored for air quality prediction.

To address these challenges, we propose a novel diffusion-driven incomplete multimodal learning approach, called *iMMAir*, to recover the missing data for robust multimodal air quality prediction. First, we design a shallow feature extractor to project raw data into the latent representation space, reconciling dimension inconsistency for subsequent recovery process. Moreover, we develop a conditional diffusion-driven recovery module that infers the distribution of missing data based on a conditional probability framework. This recovery module further incorporates a conditional enhancer to introduce temporal correlation and semantic consistency constraints, thereby mitigating the distribution gap between recovered and true data. Subsequently, we use a multimodal network to fuse recovered and available data for downstream air quality prediction tasks. Additionally, to alleviate the modality imbalance problem, we employ an adaptive gradient modulation strategy to adjust the optimization process of each modality and enhance generalization

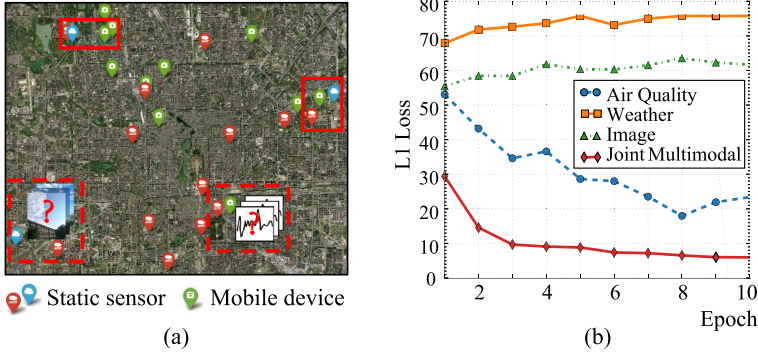


Fig. 1. (a) Distribution of air quality monitoring locations in Beijing. The solid red frames indicate areas with complete multimodal data, and the dashed red frames highlight regions with missing modalities denoted as a question mark. (b) Performance comparison of unimodal branch during the multimodal training. We can observe the multimodal model achieves the best prediction performance. However, the performance of weather and images data decreases over epochs, contrary to the improvement in overall air quality, illustrating the imbalanced optimization problem.

ability. Furthermore, we collect and curate a new multimodal dataset, named *MMAir*, composed of three heterogeneous data sources: air quality, weather, and outdoor images. In our experiments, we evaluate the performance of *iMMAir* on our curated dataset and two publicly available benchmark datasets. The results show that the proposed model outperforms the state-of-the-art baselines. The main contributions of this article are summarized as follows:

- We propose the first, to the best of our knowledge, incomplete multimodal learning approach, named *iMMAir*, for air quality prediction.
- We develop a conditional diffusion-driven recovery method to complete missing data under the constraints of semantic consistency and temporal correlation, which effectively mitigates the distribution gap between the recovered and true data.
- We employ an adaptive gradient modulation strategy to alleviate the optimization imbalance problem in multimodal learning.
- We curate the unified air quality dataset *MMAir* for evaluation. Experimental results indicate that *iMMAir* reduces prediction errors by an average of 5.6% and 2.5% in air quality regression and classification tasks, respectively.

2 Related Work and Preliminaries

In this section, we review related work on air quality prediction and multimodal representation learning, and we provide preliminaries of the fundamental concepts related to **Stochastic Differential Equation (SDE)/Ordinary Differential Equation (ODE)**-based diffusion processes.

2.1 Air Quality Prediction

Air quality prediction methods are generally divided into two categories: physics-based and data-driven methods. Traditional analytical methodologies rely on physics-based atmospheric models, demanding substantial domain expertise and involving computationally expensive processes [1, 49]. In contrast, data-driven approaches have increasingly focused on leveraging deep neural networks to capture spatiotemporal correlations using historical observations. For instance, Du et al. [9] and Athira et al. [2] have developed predictive frameworks based on recurrent neural networks to capture temporal dependencies within multisource data. In addition, spatiotemporal

deep learning methods have shown considerable adaptability in addressing the challenges of air quality prediction [10, 11, 13, 15, 44, 50]. For instance, approaches like PM_{2.5}-GNN [51] and GAGNN [6] have exploited **Graph Neural Networks (GNNs)** to facilitate effective message passing among various observation stations. In addition, Cheng et al. [7] have explored the interstation relationships using graph attention networks. Recently, transformers have demonstrated superior performance in time series analysis [61], thereby gaining popularity for air quality predicting tasks by modeling spatial and temporal dependencies [27]. However, these models rely solely on static multisensor data to analyze complex atmospheric changes, yet they still lack the capability for fine-grained, low-cost monitoring. To this end, some studies have used images to understand air pollution intuitively [4, 19, 42]. Specifically, atmospheric optics based approaches primarily extract features related to atmospheric scattering, with the dark channel prior method [19] gaining significant attention in numerous studies. In addition, deep learning based methods leverage various CNN architectures for feature extraction, such as the transfer learning based air quality estimation method [4]. Moreover, some hybrid models that integrate site data and images have emerged, thereby enhancing the overall accuracy and robustness of air quality predictions [29]. However, fusing these heterogeneous data sources depends heavily on geographic coordinates and temporal markers, often resulting in incomplete data samples. Ensuring informativeness and robustness to missing data during the test phase poses a significant challenge.

2.2 Multimodal Representation Learning

With the growing availability of diverse data sources, learning multimodal representations suitable for downstream tasks has garnered increasing attention in temporal sensor data [35, 39, 54]. Previous studies on multisource or multimodel focus on capturing spatial and temporal representations from diverse data, typically relying on concatenation-based or alignment-based fusion methods [21, 26]. For instance, Liu et al. [29] leverage CNN for images and LSTM for weather and air quality data, integrating two models' outcomes through a late fusion strategy. Zhang et al. [59] utilize multigroup feature integration and distributed fusion architecture for multisource air quality data. Recently, the attention mechanism within transformers has provided an effective method to integrate multimodal signals (e.g., [8, 54]), effectively capturing cross-modal interactions. However, challenges arise due to data collection and transmission complexity, leading to inevitable instances of missing modalities in real-world scenarios. Existing recovery methods for incomplete multimodal learning typically use deep neural networks to infer missing modalities based on the observed ones [5, 53, 60]. However, these methods fall short in scenarios involving partially missing data within individual modalities and present challenges in ensuring consistent distributions between the recovered and true data in our research. In addition, some imputing methods for air quality, such as GAN-based [33, 40] and multistep diffusion-based [47] models, are designed for unimodal data, which are typically not tailored for handling complex and diverse multimodal data like image data. Recent advances in SDE/ODE-based diffusion models have demonstrated their superior capabilities in image generation [24, 30, 45]. Inspired by this, we propose a novel diffusion-driven incomplete multimodal learning approach to recover missing data and provide robust predictions for air quality.

2.3 SDE/ODE-Based Diffusion

A diffusion model facilitates the transition from simple to complex distributions by iteratively applying noise-driven diffusion steps. In a continuous-time context, diffusion models based on the SDE or ODE provide a theoretically sound framework for generating complex data distributions from a simple distribution [20, 46, 55].

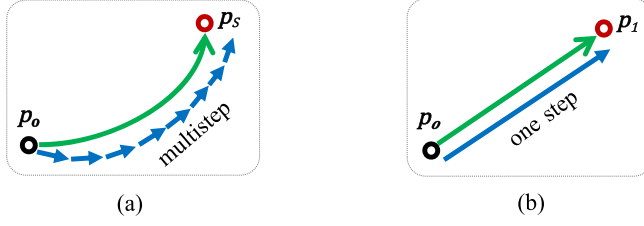


Fig. 2. Green: true SDE/ODE trajectory. Blue: discrete trajectory via numerical solvers. (a) Curved paths require finer discretization with smaller step sizes and more steps. (b) Straight paths can be simulated directly and accurately in one step.

Score-Based SDE. Given a d -dimensional data density $p_0(\mathbf{x}_0)$, noise can be injected into data at continuous-time steps $s \in [0, S]$ via the following SDE:

$$d\mathbf{x}_s = f_t(\mathbf{x}_s) ds + g_s dW_s, \quad (1)$$

where W is a standard Wiener process, the drift coefficient $f_s(\cdot) : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^d$ is a vector-valued function, and the diffusion coefficient $g_s : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar-valued function. This density evolution process can be reversed from T to 0 with a reversed SDE:

$$d\mathbf{x}_s = [f_s(\mathbf{x}_s) - g_t^2 \nabla_{\mathbf{x}_s} \log p_t(\mathbf{x}_s)] ds + g_s d\bar{W}_s, \quad (2)$$

where $\bar{W}$ represents a Wiener process in reverse time. $\nabla_{\mathbf{x}_s} \log p_s(\mathbf{x}_s)$ is the score function, which is typically parameterized by a neural network. Song and Ermon [45] have proposed an NCSN (Noise Conditional Score Network), selecting N noise levels $\{\sigma_i\}_{i=1}^N$ to perturbed data distribution—that is, $p_\sigma(\mathbf{x}_s) = \mathcal{N}(\mathbf{x}_{s-1}, \sigma^2 \mathbf{I})$. The score network $s_\theta(\mathbf{x}_s, \sigma)$ is trained using a weighted sum of score matching objectives [20].

ODE with Rectified Flow. However, the preceding diffusion process requires a tradeoff between computational efficiency and sample quality, as it introduces superfluous curvature to the generative paths from the perspective of flow, as shown in Figure 2(a). Recently, the work of Liu [30] and Liu et al. [31] presented rectified flow, which produces nearly straight flows by applying one-step diffusion and yields good performance. Specifically, given two d -dimensional data densities $p_0(\mathbf{x}_0)$ and $p_1(\mathbf{x}_1)$, the rectified flow induced by coupling $(\mathbf{x}_0, \mathbf{x}_1)$ over time $s \in [0, 1]$ is resolved by the following ODE:

$$d\mathbf{x}_s = v_s(\mathbf{x}_s, s) ds, \quad (3)$$

where the drift coefficient $v_s(\cdot, s) : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ is a time-dependent velocity function to drive the evolution trajectory to achieve the gradual transformation from p_0 to p_1 . Liu et al. [31] minimize the trajectory curvature of $\mathbf{x}_s$ by solving a simple least squares regression problem:

$$\min_s \int_0^1 \mathbb{E} [\|(\mathbf{x}_1 - \mathbf{x}_0) - v_s(\mathbf{x}_s, s)\|^2] ds, \quad (4)$$

with $\mathbf{x}_s = s\mathbf{x}_1 + (1-s)\mathbf{x}_0$,

where v_s can be parameterized by training a score-based neural network to guide the ODE to closely follow a straight path. Here, the backward process, equivalent to Equation (4) by exchanging $\mathbf{x}_0$ and $\mathbf{x}_1$, is a time reversal of the forward process. Therefore, the training algorithm treats the curvature of both trajectories equally, enabling the numerical solver to achieve flow straightening (i.e., zero curvature) in one step and produce accurate results, as shown in Figure 2(b).

In this article, we focus on ODE-based diffusion models with rectified flow to recover missing data for air quality prediction, which achieve identical marginal distributions as SDEs and offer conceptual simplicity and enhanced sampling efficiency.

3 MMAir Dataset

To comprehensively explore the air quality prediction problem, we collect and curate a unified air quality dataset, named *MMAir*, collected from different monitoring locations over 2 years. Our *MMAir* dataset comprises three distinct data sources: air quality, weather, and outdoor images. In this article, we clarify “modalities” as different data sources used, which enables us to handle various patterns of data incompleteness effectively.

Air Quality. We adopt a web crawler to gather air quality data¹ from Beijing, China, covering the period from January 1, 2015, to December 31, 2016. This dataset includes observations from 11 ground-based monitoring stations, each providing hourly measurements of six air quality indices: PM_{2.5}, PM₁₀, O₃, NO₂, CO, and SO₂.

Weather. The weather data for Beijing, China, is obtained from the National Climatic Data Center,² spanning from January 1, 2015, to December 31, 2016. This dataset includes key meteorological variables such as temperature, humidity, wind speed, and wind direction. Recognizing the influence of wind characteristics on particle transport, we further decompose the wind speed and direction into wind *u*-component and *v*-component.

Outdoor Image. Our outdoor image dataset is derived from two different sources. The first subset originates from smartphones around eight fixed monitoring sites [29], containing a series of outdoor environmental photos taken by mobile phone users between May 1, 2015, and May 1, 2016. To increase the diversity and volume of our image data, we further acquire a collection of outdoor photographs via a web crawler from Beijing tourism website,³ extending the range from January 1, 2015, to December 31, 2016. In the current setup, the image data is collected in close proximity to the air quality stations to ensure alignment with the ground-truth measurements, which helps to maintain consistency in our experiments. In future work, we plan to explore strategies for leveraging more widely distributed outdoor images, possibly by combining them with satellite data or using spatial interpolation techniques to extend the current setup. This could help enhance the spatial resolution without compromising the alignment with ground truth data.

For each set of multivariate, irregularly sampled data, we align the timestamps with a margin of half-hour error and ensure that images captured within a 2 km radius and weather data from the nearest station are matched to air quality monitoring sites. Due to the spatiotemporal constraints in our alignment procedure, the processed dataset frequently exhibits data noise and incompleteness issues. In this study, we employ linear interpolation to impute missing data within 3 hours, and continuous missing data spanning over 3 hours is marked as ‘NaN’ [18]. This preprocessing is based on the assumption that weather conditions tend to remain relatively stable over short periods, meaning that the errors introduced by this interpolation are small or controllable. For longer periods of missing data, we leave the gaps unfilled and incorporate them as part of the test data, allowing our proposed incomplete multimodal learning approach to handle the data recovery and validate its performance. In addition, our *MMAir* is labeled into six classes according to AQI (Air Quality Index) standards, which reflects real PM_{2.5} measurements, as illustrated in Figure 3(a). Examples of representative outdoor images for each label are presented in Figure 3(b).

¹<https://www.bjmemc.com.cn>

²<https://www.ncei.noaa.gov/>

³<https://www.tour-beijing.com/>

Target label: PM _{2.5} (μg/cm ³)					
0	1	2	3	4	5
≤ 50	51-100	101-150	151-200	201-300	>300

(a) Samples labeling according to AQI standards.



(b) Outdoor image examples of six labels in different locations.

Fig. 3. Classification and visualization of the MMAir dataset.

4 Proposed Method: iMMAir

4.1 Overview

Problem Formulation. Given a historical time window of length T , air quality data is represented as $\mathcal{X}^a = [x_{t-T}^a, \dots, x_t^a, \dots, x_{t-1}^a] \in \mathbb{R}^{T \times c_a}$, where x_i^a denotes the air quality indices at time i , and c_a is the number of these indices. Weather data is denoted as $\mathcal{X}^b = [x_{t-T}^b, \dots, x_t^b, \dots, x_{t-1}^b] \in \mathbb{R}^{T \times c_b}$, where c_b represents the number of distinct meteorological variables. Outdoor-image data is represented by $\mathcal{X}^o = [x_t^o] \in \mathbb{R}^{h \times w \times c_o}$, with c_o indicating the number of input channels and h, w specifying the image size. Our goal is to predict the PM_{2.5} concentrations (regression task) and the AQI standards (classification task) at time t based on the historical multimodal inputs. In incomplete multimodal learning scenarios, we employ binary masking (explained in Section 5.1) to simulate partial missing patterns. For a given modality m , we denote an observation mask as $\Delta^m = \{\delta_{1:T}^m\} \in \{0, 1\}^T$, where $\delta_t^m = 0$ indicates that x_t^m is missing, and $\delta_t^m = 1$ indicates that x_t^m is observed. Consequently, the m -modality data $\mathcal{X}^m$ is divided into two parts: the observed data $\mathbf{x}^{m,c}$ and the missing ones $\mathbf{x}^{m,i}$. We use $\mathcal{I}_i$ and $\mathcal{I}_c$ to represent the incomplete and complete modal sets, respectively. For robust air quality prediction, our approach will reconstruct the missing data $\mathbf{x}^{m,i}$ using the available information $[\mathcal{X}^{m'}; \mathbf{x}^{m,c}]$ ($m' \in \mathcal{I}_c$), which remains effective even when the entire modal data is missing.

Framework. The overall framework of our iMMAir is depicted as Figure 4, which consists of three modules: shallow feature extractor, conditional diffusion-driven recovery, and multimodal fusion and prediction. Here, Figure 4 takes the incomplete air quality data as an example to clearly illustrate our incomplete multimodal learning approach. First, the shallow feature extractor is designed to extract modal-specific features and reconcile the dimension inconsistency among these modalities. Next, we propose a conditional diffusion-driven recovery module to generate incomplete modalities from a conditional probability distribution constrained by temporal correlation and semantic consistency. Last, the recovered modality is integrated with the existing complete ones for enhanced multimodal learning in downstream air quality prediction tasks. Moreover, we employ an adaptive gradient modulation strategy to adjust each modality's optimization process, thus balancing multimodal learning and enhancing generalization ability.

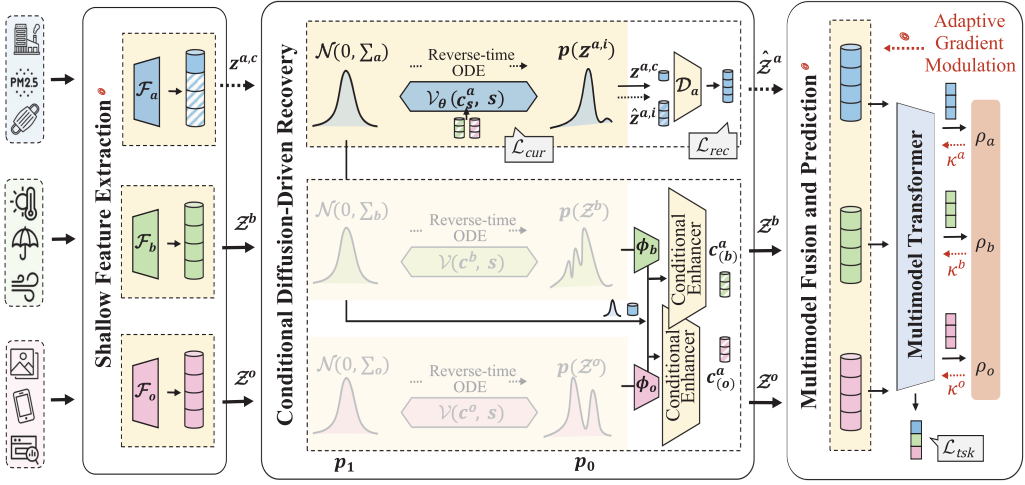


Fig. 4. Overall framework of our iMMAir. Given the input incomplete multimodal data (here we take partially missing air quality data as the example), iMMAir encodes weather data, image data, and observed air quality data to the latent vector representations $\mathcal{Z}^b, \mathcal{Z}^o, \mathcal{Z}^{a,c}$ by our shallow feature extraction module. For the missing data $\mathcal{Z}^{a,i}$, we first construct the conditional probability distribution $\mathcal{N}(0, \Sigma_a)$ based on $\mathcal{Z}^{a,c}$, which integrates available modalities $\mathcal{Z}^b, \mathcal{Z}^o$ and observed data $\mathcal{Z}^{a,c}$ as constraints to gradually reconstruct $\hat{\mathcal{Z}}^{a,i}$ in our proposed conditional diffusion-driven recovery module. Last, we fuse the recovered and available features to predict the air quality. During the training stage, an adaptive gradient modulation strategy is employed to adjust the optimization process of each modality by monitoring the discrepancy ρ_a, ρ_b, ρ_o of their contribution toward the learning objective.

4.2 Shallow Feature Extraction

Given each modality's distinct data structures and unique characteristics, we design a shallow feature extractor to capture modal-specific features effectively. To harmonize the heterogeneous data (i.e., air quality, weather, image) and facilitate their integration, this module is equipped with a separate encoder for each modality, projecting different inputs into a d -dimensional vector representation $\mathcal{Z}^m \in \mathbb{R}^d$. Thus, the process of shallow feature extraction can be expressed as follows:

$$\mathcal{Z}^m = \mathcal{F}_{\theta_1^m}^m(\mathcal{X}^m), \quad m \in (a, b, o), \quad (5)$$

where the encoded features $\mathcal{Z}^m$ are accordingly divided into two parts: the observed features $\mathcal{Z}^{m,c}$ and the missing ones $\mathcal{Z}^{m,i}$. Here, $\mathcal{F}_{\theta_1^m}^m(\cdot)$ with parameters θ_1^m encompasses two types of encoders based on the data structure:

- *Sequential encoders*: For the sequential data streams of air quality and weather data $\mathcal{X}^a, \mathcal{X}^b$, we respectively employ the single-layer LSTM networks [9, 29] to capture the temporal dependencies.
- *Visual encoder*: For $\mathcal{X}^o$, we utilize a U-Net network [4, 25] to encode outdoor-image data and extract optical information related to air pollution levels.

Here, we select the preceding encoders to stay consistent with baseline methods for a fair assessment. However, these encoders can be substituted with the corresponding established foundation models (e.g., transformers [17, 61]) to enhance potential feature extraction capabilities for specific data characteristics and structures. The subsequent modal recovery process focuses on reconstructing these shallow features of missing modalities based on the available information.

To ensure a robust representation of each modality, we further employ a two-stage training strategy. Initially, within a complete multimodal scenario, we pretrain this shallow feature extractor to discern and encode modality-specific features effectively. This pretraining stage is further enhanced by implementing unimodal loss functions L^m . It aims to minimize reconstruction errors (i.e., ℓ_2 norm), thus improving the fidelity of each modality in the latent representation space. Once the sequential and visual encoders are fine-tuned to reliably represent the intrinsic features of each modality, we freeze their parameters and then focus on optimizing the recovery and fusion module during the prediction stage.

4.3 Conditional Diffusion-Driven Recovery

To overcome the challenges posed by modal incompleteness, often due to device deployment and sensor malfunctions issues, we are committed to mitigating the impact of missing information on downstream multimodal fusion and predictive tasks. Inspired by the conditional diffusion models [47, 57], we can impute incomplete modalities using available information in an automated manner. To this end, we design a conditional diffusion-driven recovery module for modal completion to ensure robust air quality prediction even in the face of incomplete data.

4.3.1 Modal Recovery. This modal recovery process is realized using an ODE-based diffusion model with rectified flows [31], due to the efficient reconstruction capabilities that maintain the consistency of data distribution. Specifically, we infer the distribution of missing data, p_0 , from a latent distribution p_1 conditioned on the observed data. For the missing part of the m -modal features $z_0^{m,i} \sim p_0$, the diffusion model describes an evolution process: $z_0^{m,i} \mapsto z_1^{m,i}$, where $z_1^{m,i} \sim p_1$. To simplify the complex dynamics typical of the diffusion process, the trajectory of an intermediate sample $z_s^{m,i}$ in rectified flow is defined via linear interpolation:

$$z_s^{m,i} = (1-s)z_0^{m,i} + sz_1^{m,i}, \quad \text{with } s \in [0, 1], \quad (6)$$

where the latent state $z_0^{m,i}$ can be sampled from predefined distribution such as standard normal distribution [45, 47]. In our air quality prediction tasks, however, the initial state $z_0^{m,i}$ is not allowed to be observed in our air quality prediction tasks. To this end, instead of directly predicting missing features $z_0^{m,i}$, this modal recovery focuses on learning a velocity function $v(\cdot, s)$. This function drives the evolution trajectory, facilitating a gradual transformation from the initial distribution $p_0(z_0^{m,i})$ to the target $p_1(z_1^{m,i})$. This transfer process can be achieved via the following ODE:

$$\frac{dz_s^{m,i}}{ds} = v(z_s^{m,i}, s), \quad (7)$$

where $v(\cdot, s)$ can be approximate by a well-trained score-based neural network $\mathcal{V}_{\theta_2^m}(z_s^{m,i}, s)$ parameterized by θ_2^m . This network ensures that the transfer process between $z_0^{m,i}$ and $z_1^{m,i}$ closely follows a straight trajectory. Therefore, we constrain the deviation of trajectory curvature by minimizing the matching loss function:

$$\mathcal{L}_{cur} = \mathbb{E}_{m \in \mathcal{I}_i} \left\| (z_1^{m,i} - z_0^{m,i}) - \mathcal{V}_{\theta_2^m}(z_s^{m,i}, s) \right\|^2. \quad (8)$$

After training this score-based network $\mathcal{V}_{\theta_2^m}(z_s^{m,i}, s)$, we employ a reverse-time ODE to obtain the estimated distribution p_0 of the missing data from the latent distribution p_1 . Specifically, we implement the inverse dynamics by exchanging $z_0^{m,i}$ and $z_1^{m,i}$ in Equation (7)—that is, $d\hat{z}_s^{m,i} = -\mathcal{V}_{\theta_2^m}(\hat{z}_s^{m,i}, s)ds$, starting from $\hat{z}_0^{m,i}$ drawn from p_1 . Typically, generative models begin with a standard normal distribution p_1 . While this method has shown practical effectiveness, it often fails to capture the unique and complex characteristics inherent in specific modality, thereby increasing the difficulty of distribution transformation process. Therefore, the lack of explicit guidance and

constraints often leads to omitting crucial information, weakening the ability to capture complex intrasequence and intermodal dependencies in the modal recovery process.

4.3.2 Conditional Constraints. To effectively capture the intrasequence and intermodal dependencies based on different shallow features, iMMAir designs a conditional diffusion-driven recovery module to facilitate a precise transformation from p_1 to p_0 . Specifically, we integrate two pivotal conditional constraints into the modal recovery process: the temporal correlation of each modality and the semantic consistency across different modalities. Introducing these constraints enables a robust recovery of missing data while preserving the essential characteristics inherent to each modality.

Temporal Correlation. To bridge the distribution gap between observed and missing data, we enhance the latent distribution p_1 by integrating prior knowledge. Rather than relying solely on the standard normal distribution, we employ a Gaussian process to introduce the temporal constraints within the sequence, thus modeling the latent distribution accurately [47, 52]. Given the disparities across intermodal distributions caused by different data types, we formulate this Gaussian process using the observed parts of intramodal data. Specifically, we construct a joint distribution for the missing $z^{m,i}$ and observed data $z^{m,c}$ within the latent space, which is articulated as follows:

$$\begin{bmatrix} z^{m,i} \\ z^{m,c} \end{bmatrix} \sim \mathcal{N}\left(\mu_m, \begin{bmatrix} \Sigma_{ii} & \Sigma_{ic} \\ \Sigma_{ci} & \Sigma_{cc} \end{bmatrix}\right), \quad (9)$$

where the mean μ_m is set to zero for simplicity. The kernel function Σ_m , serving as the covariance matrix, is employed to capture the temporal correlation between $z^{m,i}$ and $z^{m,c}$. Consequently, the conditional distribution $p(z^{m,i}|z^{m,c})$ can be imputed as follows:

$$p(z^{m,i}|z^{m,c}) = \mathcal{N}(\Sigma_{ic}\Sigma_{cc}^{-1}z^{m,c}, \Sigma_{ii} - \Sigma_{ic}\Sigma_{cc}^{-1}\Sigma_{ci}). \quad (10)$$

To avoid this expensive cubic operation, we further adopt a kernel assumption that renders the computation linear in time [3]. In our work, $p(z^{m,i}|z^{m,c})$ is employed as the prior distribution for missing data within the latent space. Therefore, instead of generating random noise from the standard normal distribution $\mathcal{N}(0, \mathbf{I})$, we draw samples $z_1^{m,i}$ from a conditional distribution $\mathcal{N}(0, \Sigma_m)$ as defined in the conditional framework of Equation (10), to enhance temporal constraints for effective modal recovery. Notably, in scenarios where an entire modality is missing, iMMAir defaults to sampling from a standard normal distribution, thus maintaining the integrity of our recovery task.

Semantic Consistency. Considering the semantic consistency across various modalities within identical spatiotemporal contexts, we further enhance our modal recovery by incorporate the existing complete modalities to infer missing m -modality—that is, $\mathcal{Z}^m \leftarrow \{\phi_{m'}(\mathcal{Z}^{m'}) | m' \in \mathcal{I}_c\}$. Specifically, we employ a 1D convolution operation $\phi(\cdot)$ to harmonize different dimensions and extract relevant features. Therefore, the semantic information of the corresponding available modalities are integrated in the missing parts $z^{m,i}$ of the incomplete modalities.

Subsequently, we integrate two constraints into the final diffusion-driven recovery process by a conditional enhancer: $c_{(m')}^m = C_m(z_1^{m,i}, z^{m,c}, \phi_{m'}(\mathcal{Z}^{m'}))$, where $z_1^{m,i} \sim p(z^{m,i}|z^{m,c})$. Here, this enhancer utilizes convolutional transformations for feature concatenation and alignment, thereby seamlessly incorporating temporal and semantic constraints for effective modal recovery. In our work, we employ NCSNs [45, 46] to obtain the score of distribution estimate and guide the recovery process. Utilizing these obtained scores, iMMAir then minimizes the new trajectory curvature for our conditional diffusion-driven recovery learning, which can be formulated as follows:

$$\begin{aligned} \mathcal{L}_{cur} &= \mathbb{E}_{m \in \mathcal{I}_i} \left\| (c^m - z^{m,i}) - \mathcal{V}_{\theta_2^m}(z_s^{m,i}, s) \right\|^2, \\ &\quad \text{with } z_s^{m,i} = (1-s)c^m + sz^{m,i}. \end{aligned} \quad (11)$$

Therefore, it becomes intuitive to seek a coupling $(\mathbf{c}^m, \mathbf{z}^{m,i})$ that minimizes the matching loss $\mathcal{L}_{cur}$. In the inference phase, we aim to approximate $\mathbf{z}^{m,i}$ with $s = 0$ using the well-trained score-based network via a one-step diffusion process—that is, $\hat{\mathbf{z}}^{m,i} = \mathbf{c}^m + \mathcal{V}_{\theta_2^m}(\mathbf{c}^m, 0)$. In addition, to improve the estimation precision with high variability in the inference process, we further introduce a lightweight decoder $\mathcal{D}_m(\cdot)$, which refines the recovered representations for incomplete modalities. To this end, we further compute the reconstruction loss $\mathcal{L}_{rec}$ between the original and recovered features for the missing part, which is formulated as follows:

$$\mathcal{L}_{rec} = \mathbb{E}_{m \in \mathcal{I}_i} \left\| \mathcal{D}_m(\hat{\mathbf{z}}^{m,i}) - \mathbf{z}^{m,i} \right\|^2, \quad (12)$$

where the decoder $\mathcal{D}_m(\cdot)$ consists of residual channel-attention blocks [58] and 1D convolutional layers to refine recovered features. Finally, we complete the missing features for subsequent multimodal fusion and prediction tasks: $\hat{\mathbf{Z}}^m = [\mathcal{D}_m(\hat{\mathbf{z}}^{m,i}); \mathbf{z}^{m,c}]$. Notably, our recovery module is designed to be highly plug-and-play, enabling effortless adaptation to various missing data scenarios and facilitating compatibility with other predictive models without extra processing. The constraints can be extended by introducing spatial and temporal correlations, allowing the model to handle cases where data are collected from distant locations within the same time period or from the same location at different times.

4.4 Multimodal Fusion and Prediction

The primary goal of iMMAir is to recover missing data, thereby facilitating complete multimodal fusion for robust air quality prediction. As depicted in Figure 4, we utilize a multimodal transformer architecture $\mathcal{T}_{\theta_3}(\cdot)$ parameterized by θ_3 with a cross-attention mechanism [54], thus allowing iMMAir to facilitate effective fusion and interaction across different modalities. Additionally, we implement attention mechanisms and feed-forward networks on inverted dimensions [32], which helps reveal complex intermodal correlations and capture intramodal temporal dependencies. To ensure optimal performance of iMMAir, we integrate three loss functions in the prediction phase, collectively forming our final optimization objective:

$$\mathcal{L} = \mathcal{L}_{tsk} + \lambda(\mathcal{L}_{cur} + \mathcal{L}_{rec}), \quad (13)$$

where $\mathcal{L}_{cur}$ and $\mathcal{L}_{rec}$ represent the averaged matching and reconstruction losses, respectively, focusing on the incomplete modality learning. The task-specific loss, $\mathcal{L}_{tsk}$, is tailored to the predictive goals of the model, employing **Mean Absolute Error (MAE)** for regression tasks and cross-entropy loss for classification tasks. The coefficient λ is designed to adjust the relative weighting between recovery and prediction efforts, ensuring an optimal tradeoff for overall model performance.

However, integrating multiple modalities often leads to inefficiencies, as the dominant modality may overshadow the contributions of weaker ones, thereby preventing our model from fully exploiting all available information. To address this issue, we implement an adaptive gradient modulation strategy inspired by the On-the-Fly Gradient Modulation Generalization Enhancement algorithm [38]. Specifically, we monitor the contribution scores of each modality, $u_m, m \in (a, b, o)$, according to the unimodal's learning objective (e.g., L^m). We then compute the relative discrepancies ρ_m , such as $\rho_a = u_a / \max(u_b, u_o)$, which helps us identify modalities that are underperforming. During training, our iMMAir incorporates an adaptive strategy to modulate the gradient ∇_{θ^m} . The parameters $\theta_{(i)}^m$ at the i -iteration are updated as follows:

$$\begin{aligned} \theta_{(i+1)}^m &= \theta_{(i)}^m - \alpha \kappa_{(i)}^m \nabla_{\theta^m} L^m(\theta_{(i)}^m), \\ \kappa^m &= \begin{cases} 1 - \tanh(\beta \rho_m) & \text{if } \rho_m > 1, \\ 1 & \text{otherwise,} \end{cases} \end{aligned} \quad (14)$$

ALGORITHM 1: Training Process for iMMAir

Require: Pretrained available features $\mathbf{z}^{a,c}$, $\mathcal{Z}^b$, $\mathcal{Z}^o$, recovered features $\mathbf{z}^{a,i}$, predictive target $\mathbf{y}$, hyperparameter λ , η ; learning rate α , iteration number I .

Ensure: Optimized parameters for iMMAir.

1: **for** $i = 0, \dots, I - 1$ **do**

2: Construct $\mathcal{N}(0, \Sigma_a)$ through Gaussian process and sample $\mathbf{z}_1^{a,i}$ via Equation (10);

3: Integrate constraints: $\mathbf{c}_{(b)}^a = C_a(\mathbf{z}_1^{a,i}, \mathbf{z}^{a,c}, \phi_b(\mathcal{Z}^b))$, $\mathbf{c}_{(o)}^a = C_a(\mathbf{z}_1^{a,i}, \mathbf{z}^{a,c}, \phi_o(\mathcal{Z}^o))$;

4: Randomly select step: $s \in [0, 1]$ and compute the matching loss $\mathcal{L}_{cur}$ via Equation (11);

5: Recover $\hat{\mathbf{z}}^{a,i}$ using concatenation operation:

$$\hat{\mathbf{z}}^{a,i} \leftarrow [\mathbf{c}_{(b)}^a + \mathcal{V}_{\theta_2}(\mathbf{c}_{(b)}^a, 0) \parallel \mathbf{c}_{(o)}^a + \mathcal{V}_{\theta_2}(\mathbf{c}_{(o)}^a, 0)];$$

6: Complete the features: $\hat{\mathcal{Z}}^a \leftarrow [\mathcal{D}_a(\hat{\mathbf{z}}^{a,i}); \mathbf{z}^{a,c}]$;

7: Compute the reconstruction loss $\mathcal{L}_{rec}$ via Equation (12);

8: Multimodal fusion and prediction:

$$\mathcal{T}_{\theta_3}(\hat{\mathcal{Z}}^a, \mathcal{Z}^b, \mathcal{Z}^o) \rightarrow \hat{\mathbf{y}}, \hat{\mathbf{y}}_a, \hat{\mathbf{y}}_b, \hat{\mathbf{y}}_o;$$

9: Calculate the discrepancies: $\rho_a, \rho_b, \rho_o \leftarrow \hat{\mathbf{y}}_a, \hat{\mathbf{y}}_b, \hat{\mathbf{y}}_o$;

10: Compute the task-specific loss $\mathcal{L}_{tsk}$ and integrate the total loss $\mathcal{L}$ via Equation (13);

11: Adjust the gradient of each modality during the optimization process and update model parameters in iteration i using the Adam optimizer via Equation (14).

12: **end for**

where α denotes the learning rate, and κ^m serves as an adaptive modulation coefficient, which is integrated into the optimization process to adjust the gradient for each modality. The hyperparameter β controls the intensity of this modulation, enabling κ^m to selectively attract more attention to the underoptimized modalities while leaving other modalities unaffected. To better focus on the reconstruction of important modalities, we limit the gradient modulation to shallow feature extraction (θ_1) and multimodal prediction (θ_3) modules. Therefore, our adaptive modulation strategy effectively optimizes the contribution of each modality to the overall model performance, ensuring a balanced training process. The entire optimization process for model training is executed in an end-to-end manner, as meticulously outlined in Algorithm 1, with the validation processes explicitly detailed in Algorithm 2. Here, we describe two processes to simulate the incomplete case of in one modality (i.e., air quality) using a random missingness strategy. We assume that our shallow feature extraction module has been pretrained via Equation (5) on a complete multimodal dataset.

5 Experiments and Analysis

In this section, we demonstrate the effectiveness of iMMAir for air quality prediction tasks, and conduct experiments on our curated *MMAir* dataset and two public benchmark datasets for comprehensive evaluation.

5.1 Implementation

Dataset Settings. To evaluate the effectiveness of incomplete multimodal learning for air quality prediction, we adopt a random missing strategy to simulate the absence of modal samples. Here,

ALGORITHM 2: Validation Process of iMMAir

Require: Pretrained available features $\mathbf{z}^{a,c}, \mathcal{Z}^b, \mathcal{Z}^o$; iMMAir with optimized parameters.

Ensure: Predictive target $\hat{\mathbf{y}}$.

- 1: Construct $\mathcal{N}(0, \Sigma_a)$ and sample $\mathbf{z}_1^{a,i}$;
- 2: Integrate constraints: $\mathbf{c}_{(b)}^a = C_a(\mathbf{z}_1^{a,i}, \mathbf{z}^{a,c}, \phi_b(\mathcal{Z}^b))$, $\mathbf{c}_{(o)}^a = C_a(\mathbf{z}_1^{a,i}, \mathbf{z}^{a,c}, \phi_o(\mathcal{Z}^o))$;
- 3: Recover $\hat{\mathbf{z}}^{a,i}$ with $s = 0$ using concatenation operation:

$$\hat{\mathbf{z}}^{a,i} \leftarrow [\mathbf{c}_{(b)}^a + \mathcal{V}_{\theta_b^b}(\mathbf{c}_{(b)}^a, 0) \parallel \mathbf{c}_{(o)}^a + \mathcal{V}_{\theta_o^o}(\mathbf{c}_{(o)}^a, 0)];$$

- 4: Complete the features: $\hat{\mathcal{Z}}^a \leftarrow [\mathcal{D}_a(\hat{\mathbf{z}}^{a,i}); \mathbf{z}^{a,c}]$;
- 5: Multimodal fusion and prediction: $\mathcal{T}_{\theta_3}(\hat{\mathcal{Z}}^a, \mathcal{Z}^b, \mathcal{Z}^o) \rightarrow \hat{\mathbf{y}}$.

our MMAir dataset comprises three distinct modalities and exhibits seven different patterns of incompleteness. Specifically, we define a missing rate, η , and randomly remove k modalities from the original N complete samples until the condition $\eta = \sum_{i=1}^N k_i / 3N$ is achieved. In addition, we utilize a binary masking strategy to simulate structured missing patterns [47] for air quality and weather data, treating entire modalities as completely missing during full-sequence masking. Conversely, the image data does not follow this masking strategy since it is either fully available or entirely absent, without partial missing scenarios. In our experiments, we meticulously ensure the presence of at least one modality by selecting η from the set $[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7]$, where 0.7 represents the scenario with approximately one modality available in all data samples. We then maintain these missing rates consistently across the training, validation, and testing phases. Furthermore, we distribute the samples across all datasets in 7:1:2 for training, validation, and testing.

Training Details. For a fair comparison, we conduct a comprehensive grid search for all methods to determine the optimal combination of hyperparameters. We use the Adam optimizer during the training phase, starting with an initial learning rate of 0.0001. To adjust the learning rate based on the model's performance, we further utilize the ReduceLROnPlateau scheduler with a patience of 5 epochs and a decay factor of 0.5. In addition, the batch size is set at 128, the window size at 48, and the hidden feature dimension at 128. The modulation parameter β is determined from $[0, 1]$ according to the validation set. We set the hyperparameter λ at 10 for the regression task and 1 for the classification task, which will be further discussed in Section 5.5. To ensure the reliability and reproducibility of our results, each experiment is replicated five times, with the average outcomes reported on the testing set. All computational experiments are conducted on a machine with four NVIDIA P100 GPUs.

Evaluation Metrics. For a comprehensive analysis, we assess the performance of the regression models by calculating the MAE and the **Root Mean Square Error (RMSE)**. Additionally, the performance of the classification models is evaluated using the Accuracy and Macro-F1 Score metrics. We performed significant tests using the paired t -test. Differences are considered statistically significant when the p -value is lower than 0.05.

5.2 Performance Comparison

Since the incomplete multimodal learning for air quality prediction has yet to be well explored, we compare our iMMAir model with the following four baseline models. These models have been appropriately modified to suit our specific tasks. They include two methods that are focused on air quality analysis and two that are based on established multimodal learning methods:

- *Multi-LSTM* [41] employs a unified LSTM architecture to capture temporal dependencies both within individual modalities and across different modalities, facilitating late fusion for label integration.
- *Third-Eye* [29] combines CNNs, employing a dark channel prior for image processing, with recurrent neural networks for handling air quality and weather data, culminating in a late weighted fusion process for feature integration.
- *Ensemble* [4] utilizes an ensemble strategy of four feature extractors specifically for raw image data, and subsequently integrates our pretrained air quality and weather features through late fusion.
- *DiCMoR* [53] implements a normalizing flow-based modality recovery technique that transfers distributions from available modalities to missing ones. For a balanced comparison, we replace only the recovery module in our proposed model with DiCMoR, thus maintaining equivalent performance in complete multimodal scenarios.

Table 1 summarizes the experimental results for various models applied to air quality regression and classification tasks under different missing rates (η). We observed that our proposed iMMAir model outperforms all compared methods, showing an average performance enhancement of approximately 5.6% in regression tasks and 2.5% in classification tasks.

In the context of complete multimodal learning, the Multi-LSTM model demonstrates limited effectiveness, primarily due to its simplistic approach to fusing sequence features from diverse modalities. In addition, we can find that models such as Third-Eye and Ensemble, which integrate visual feature extraction of air pollution levels, exhibit superior performance in air quality prediction tasks. These findings confirm that enhancing the representational capacity of outdoor images markedly boosts prediction accuracy. Additionally, our approach not only captures sequential and visual information in our shallow feature extractor but also probes modality interaction and fusion through a subsequent multimodal transformer, achieving state-of-the-art performance in feature fusion. During the optimization phase, our iMMAir model adopts an adaptive gradient modulation strategy to handle the modality imbalance problem, thus putting more effort into the underoptimized modalities.

In scenarios of incomplete multimodal learning with different missing rates (η), our approach consistently outperforms other baseline methods. Unlike Multi-LSTM and Third-Eye models that rely on basic imputation techniques such as zero or mean filling for missing data, iMMAir incorporates an additional recovery module, resulting in demonstrably superior performance. Furthermore, our method significantly outperforms the current state of the art in modal recovery approaches, DiCMoR. This highlights the superior capability of diffusion-driven recovery to handle the complexities inherent in multimodal data distributions, particularly with unstructured data like images, which will be discussed in Section 5.4. Additionally, we introduce two conditional constraints to further enhance the data reconstruction capabilities for missing data. Last, our iMMAir is adept at handling both partial and entire modal absences, making it exceptionally suited for applications in air quality monitoring.

5.3 Ablation Study

In this section, we conduct extensive ablative experiments on several model variants to verify the effectiveness of two key components in our proposed iMMAir.

5.3.1 Effects of Conditional Distribution. Recall that our iMMAir incorporates a Gaussian process to model the conditional probability distribution $\mathcal{N}(0, \Sigma_m)$ based on sequential observations, enhancing the temporal correlation within data modalities. Our conditional diffusion-driven recovery module replaces the standard normal distribution $\mathcal{N}(0, \mathbf{I})$, commonly used as the latent

Table 1. Performance Comparison of iMMAir and Baselines with Different Missing Rates η

Task	η	Multi-LSTM	Third-Eye	Ensemble	DiCMoR	iMMAir(w/o)	iMMAir
Regression	0.0	14.77/26.32	13.65/23.76	13.40/23.90	–	–	13.01/21.85
	0.1	17.22/29.57	16.91/28.47	16.74/28.24	16.33/27.16	<u>16.23/26.03</u>	15.51/25.60
	0.2	19.45/34.14	18.93/32.61	18.25/32.67	17.35/30.29	16.54/27.84	<u>16.96/28.06</u>
	0.3	21.43/36.22	19.69/34.95	19.13/34.70	18.16/31.76	<u>17.80/29.02</u>	17.66/28.41
	0.4	23.93/40.48	21.73/36.02	21.84/36.04	20.85/32.67	<u>19.97/30.18</u>	18.16/29.07
	0.5	24.81/43.36	22.73/39.65	22.52/39.30	21.24/34.12	<u>20.48/31.24</u>	18.46/30.22
	0.6	25.59/45.48	23.77/44.15	23.25/43.06	22.10/37.82	<u>20.94/33.77</u>	19.34/32.74
	0.7	26.14/48.09	24.44/46.76	24.25/46.60	23.00/38.88	<u>21.79/35.24</u>	21.01/34.88
	Avg	21.67/37.96	20.23/35.79	19.92/35.56	19.01/30.56	<u>18.35/29.39</u>	17.51/28.86
Classification	0.0	86.64/76.87	87.30/77.67	88.74/78.75	–	–	90.09/81.43
	0.1	84.95/70.91	85.28/74.20	86.28/75.50	88.65/79.70	<u>88.82/80.68</u>	89.01/81.03
	0.2	83.28/68.75	83.12/72.09	83.91/73.32	87.10/74.68	<u>88.06/77.17</u>	88.48/77.98
	0.3	81.41/65.56	82.31/69.17	82.22/70.23	86.79/73.46	<u>87.26/75.97</u>	87.74/76.43
	0.4	80.15/64.19	81.73/68.13	81.80/68.53	85.44/73.14	86.79/75.16	<u>86.46/74.39</u>
	0.5	79.41/63.76	80.99/65.92	80.95/67.57	84.52/72.57	<u>85.27/73.96</u>	86.09/74.13
	0.6	78.93/62.83	79.54/64.98	80.02/65.30	83.00/71.21	<u>84.34/73.08</u>	85.10/73.82
	0.7	77.47/62.49	79.00/63.92	79.93/64.98	82.47/69.46	<u>83.12/69.97</u>	83.89/71.22
	Avg	81.53/66.92	82.41/69.51	82.98/70.52	86.01/74.46	<u>86.72/75.93</u>	87.11/76.30

The values reported in each cell denote MAE/RMSE for the regression task and Accuracy/Macro-F1 for the classification task. iMMAir(w/o) denotes using the standard normal distribution. The ‘Avg’ row represents the average performance across different missing rates for each method. Bold and underlined values show the best and second-best values, respectively.

distribution in diffusion models. To evaluate the effectiveness of this approach, we compare our model to a variant, iMMAir(w/o), which omits the Gaussian process and instead relies on the standard normal distribution directly. As shown in Table 1, our findings indicate that employing a conditional distribution markedly enhances performance across nearly all metrics for both regression and classification tasks. In contrast, the iMMAir(w/o) variant is vulnerable to an inadequate diffusion process, resulting in less accurate distribution transformations and increased inference errors. This comparative analysis aligns with our intuition that introducing conditional constraints into the prior distribution may enhance sequential correlation and straighten the evolution trajectory, thus significantly optimizing the performance of the diffusion process. However, the complete absence of a specific modality or the emergence of extreme weather conditions may affect the construction of conditional probability distributions, leading to potential performance degradation. In such cases, reverting to the standard normal distribution can exploit the generative potential, providing greater robustness in forecasts.

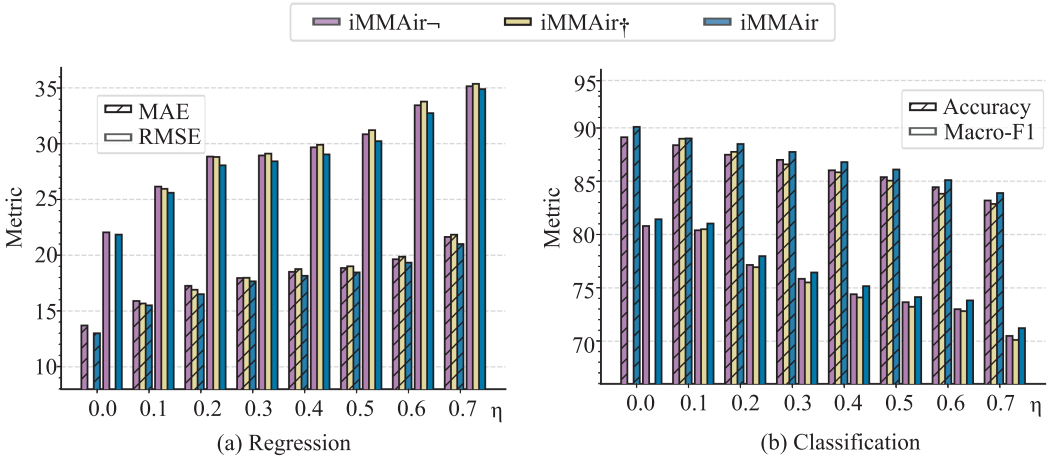


Fig. 5. Ablation comparisons of iMMAir for regression and classification tasks of air quality prediction.

5.3.2 Effects of the Adaptive Gradient Modulation Strategy. In the shallow feature extraction and multimodal prediction modules of our proposed iMMAir, we employ an adaptive gradient modulation strategy to address the optimization imbalances across different modalities. To evaluate this strategy's effectiveness, we further conduct two distinct experiments: one with iMMAir trained without this strategy (denoted as iMMAir₋) and another where the strategy is applied across all modules (denoted as iMMAir₊). The comparative results of these experiments are presented in Figure 5. It is clear that our proposed iMMAir consistently outperforms its variants across air quality regression (a) and classification (b) tasks under varying missing data rates (η). Specifically, we observe that iMMAir₋ exhibits suboptimal performance due to its inability to fully leverage and balance the contributions of different modalities. In addition, while iMMAir₊ is designed to optimize all modules, it still suffers performance degradation, especially as the missing rate increases. This unexpected result suggests that the gradient modulation strategy excessively focuses on reconstructing suboptimal modalities, ultimately causing underfitting in the final prediction. In contrast, our proposed iMMAir maintains robust performance even at higher missing rates, demonstrating a solid capability to accurately recover the missing data and effectively utilize the potential representations of each modality.

5.4 Recovery Analysis

Recall that the conditional diffusion-driven recovery approach infers the distribution of missing data based on available information to mitigate the distribution gap between the recovered and true data. To this end, we evaluate the effectiveness of our iMMAir in reconstructing missing data distributions, in comparison to the second-best method, DiCMoR. To qualitatively analyze the distribution transformation of the recovery process, we visualize the distribution of both recovered data and ground truth in the feature space. Specifically, we randomly select 100 feature samples from the last layer of the recovery module and project them into a 2D space using t-SNE for clear visualization.

The comparative results of different missing data scenarios for both DiCMoR and our proposed iMMAir are depicted in Figure 6 and Figure 7. Figure 6 visualizes the distributions of recovered and true features when only one modality is available, whereas Figure 7 shows the scenario where a single modality is absent. Consistent with our previous analyses, iMMAir infers the data distribution

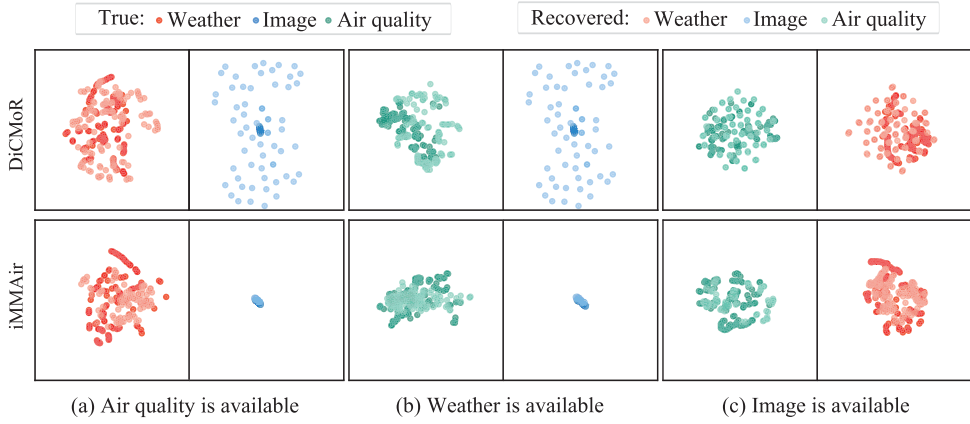


Fig. 6. Visualization of the distribution between recovered data (light color) and ground truth (dark color) for DiCMoR and our model under missing patterns with only one modality available.

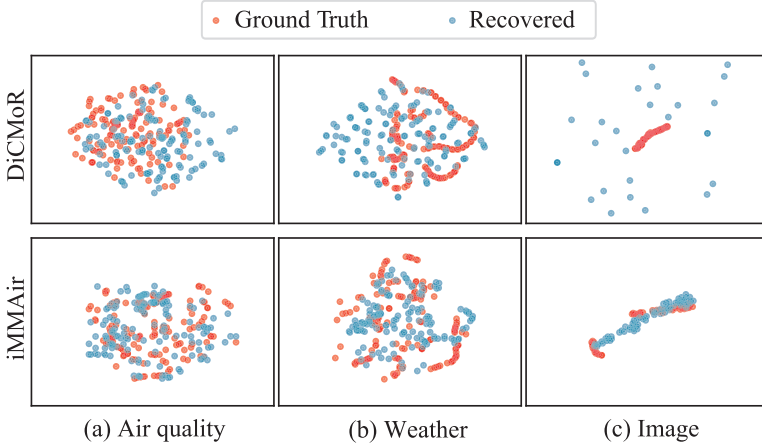


Fig. 7. Visualization of the distribution between recovered data (blue) and ground truth (red) by DiCMoR and our proposed model under missing patterns with only one modality.

that more closely aligns with the ground truth. In contrast, while DiCMoR effectively estimates the distribution transfer, it tends to exhibit significantly higher variance. Furthermore, DiCMoR faces challenges in recovering unstructured data, mainly due to the insufficient constraints. This deficiency weakens the model's generalization ability, particularly in processing visual features, showing increased dispersion and variability. iMMAir leverages a diffusion-driven approach for distribution transformation, significantly enhancing the capability to generate structured and unstructured data. Our modal recovery process not only achieves a closer approximation to the true data distribution but also effectively integrates temporal correlations and maintains semantic consistency, thus improving the overall reconstruction quality.

5.5 Hyperparameter Study

Recall that our optimization objective incorporates multiple loss functions, with the hyperparameter λ serving as a crucial factor to balance these loss terms. In this section, we explore the sensitivity

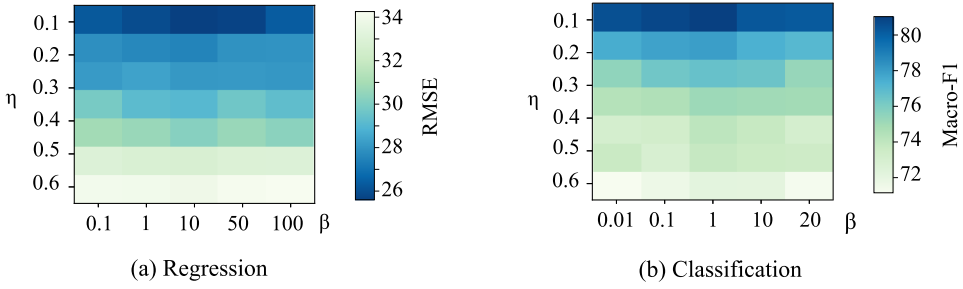


Fig. 8. Heatmaps on two air quality tasks. The x-axis denotes the weight β of loss terms in $\{0.1, 1, 10, 50, 100\}$ for the regression task or $\{0.01, 0.1, 1, 10, 20\}$ for the classification task, and the y-axis denotes different missing rates in $\{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7\}$. A darker color represents a better performance.

of λ across various missing rates η . Given the task-specific nature of λ , we select values from the sets $\{0.1, 1, 10, 50, 100\}$ for the regression task (a) and $\{0.01, 0.1, 1, 10, 20\}$ for the classification task (b), as illustrated in Figure 8. Our findings reveal that at a fixed missing rate, the performance of iMMAir generally improves with an increase in λ up to a certain optimal point. This improvement demonstrates the benefits of our model's emphasis on conditional diffusion-driven recovery, which can mitigate the distribution gap between the recovered and true data, thereby enhancing predictive accuracy. However, further increasing λ beyond this optimal point leads to a decline in performance, which may be attributed to the excessive attention to the recovery task, exacerbating the prediction difficulty. Based on these observations, we select optimal λ values with consideration of their impact on performance: $\lambda = 10$ for the regression task and $\lambda = 1$ for the classification task. This careful setting ensures that the model maintains a balanced focus on both accurate modal recovery and efficient prediction, optimizing performance across different missing data scenarios.

5.6 Validation of Benchmark Datasets

Considering the spatiotemporal unevenness and sparsity of the *MMAir* dataset, we conduct experiments on two publicly available benchmark datasets to further evaluate the effectiveness and applicability of our iMMAir. Given that high-quality annotated image data is inaccessible and challenging to collect, we simulate bimodal datasets using joint air quality and weather data, which contain 35 and 13 major monitoring stations for Beijing and London, respectively. The real-world datasets⁴ are collected from two cities, Beijing and London, which are commonly used benchmarks for air quality prediction [13, 14, 34]. Since the maximum missing rate η for bimodal datasets is 0.5, ensuring that at least one modality remains available, we select η from the set $\{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$. To verify the effectiveness of our model architecture, we compare our proposed iMMAir with other six baselines for air quality regression task under two categories:

- (1) Recovery methods, including those focused on incomplete multimodal learning (normalizing flow based [53]) and time series imputing (GAN-based [56], multistep diffusion-based [47]) methods:
 - *GAIN* [56] proposes a generative adversarial imputation network in which the generator imputes missing data while the discriminator distinguishes between observed and imputed components.

⁴https://www.biendata.xyz/competition/kdd_2018/data/

Table 2. Performance Comparison of iMMAir Versus Recovery Methods on Two Publicly Benchmark Datasets

Data	η	GAIN [56]	CSDI [47]	DiCMoR [53]	iMMAir
Beijing	0.0	6.07/11.04	–	–	–
	0.1	6.51/11.37	6.42/11.33	<u>6.39/11.30</u>	6.31/11.27
	0.2	6.64/11.66	<u>6.54/11.54</u>	6.58/11.56	6.48/11.47
	0.3	6.91/11.90	6.83/11.81	<u>6.77/11.74</u>	6.66/11.64
	0.4	7.05/12.23	6.97/12.10	<u>6.90/12.04</u>	6.82/11.89
	0.5	7.23/12.49	7.15/12.38	<u>7.07/12.27</u>	7.02/12.10
	Avg	6.87/11.93	6.78/11.83	<u>6.74/11.78</u>	6.65/11.67
London	0.0	2.23/3.20	–	–	–
	0.1	2.39/3.44	<u>2.37/3.38</u>	2.38/3.39	2.35/3.34
	0.2	2.46/3.65	2.52/3.71	<u>2.43/3.54</u>	2.39/3.51
	0.3	2.54/3.75	2.59/3.82	<u>2.50/3.69</u>	2.49/3.63
	0.4	2.61/3.81	2.59/3.82	<u>2.59/3.78</u>	2.57/3.75
	0.5	2.67/3.95	2.65/3.90	<u>2.64/3.87</u>	2.61/3.83
	Avg	2.53/3.72	2.52/3.67	<u>2.51/3.66</u>	2.48/3.61

The values reported in each cell denote MAE/RMSE. The ‘Avg’ row represents the average performance across different missing rates, excluding $\eta = 0.0$. Bold and underlined values show the best and second-best values, respectively.

- CSDI [47] introduces a score-based diffusion model that learns the conditional distribution for time series probabilistic imputation.
- DiCMoR [53] is the second-best method on our MMAir.
- (2) Spatiotemporal deep learning methods:
 - $PM_{2.5}$ -GNN [51] constructs a directed graph for $PM_{2.5}$ prediction, modeling long-term dependencies using domain knowledge.
 - GAGNN [6] introduces a group-aware GNN to model the spatial and latent dependencies between monitoring sites for air quality forecasting.
 - Airformer [27] presents a novel transformer architecture for air quality prediction, incorporating dartboard spatial and causal temporal attention mechanisms to capture spatial and temporal dependencies, respectively.

5.6.1 Validation of Recovery Methods. To evaluate the architectural effectiveness of our conditional diffusion-driven recovery approach, we replace this module with these recovery methods. The experimental results, presented in Table 2, indicate that our iMMAir consistently outperforms all baselines across different missing rates (η). Specifically, due to its adversarial nature, GAIN often struggles with training instability and thus fails to adequately focus on predictive goals, leading to poor performance. Moreover, DiCMoR exhibits diminished performance due to the lack of essential conditional constraints, which is especially noticeable at higher missing rates. In contrast, our model demonstrates superior capability in modeling complex data distributions and precisely reflecting dynamic changes. Moreover, although iMMAir’s computational cost is slightly higher than

Table 3. Performance Comparison of iMMAir Versus Spatial-Temporal Air Quality Methods on Two Publicly Benchmark Datasets

Data	η	PM _{2.5} -GNN [51]	GAGNN [6]	Airformer [27]	iMMAir
Beijing	0.0	6.23/11.21	6.16/11.12	6.02/10.98	<u>6.07/11.04</u>
	0.1	6.73/11.72	6.71/11.61	<u>6.35/11.32</u>	6.31/11.27
	0.2	7.09/12.59	7.03/12.18	<u>6.80/11.65</u>	6.48/11.47
	0.3	7.56/13.11	7.45/12.78	<u>6.77/11.74</u>	6.66/11.64
	0.4	7.88/13.31	7.72/13.07	<u>7.68/13.01</u>	6.82/11.89
	0.5	7.96/13.50	<u>7.85/13.20</u>	7.89/13.36	7.02/12.10
	Avg	7.29/12.57	7.15/12.32	<u>6.98/12.19</u>	6.57/11.57
London	0.0	2.26/3.31	2.23/3.24	2.21/3.18	<u>2.23/3.20</u>
	0.1	2.43/3.50	2.40/3.44	<u>2.36/3.36</u>	2.35/3.34
	0.2	2.62/3.73	2.55/3.69	<u>2.45/3.55</u>	2.39/3.51
	0.3	2.65/3.89	2.62/3.88	<u>2.61/3.86</u>	2.49/3.63
	0.4	2.70/4.02	<u>2.63/3.93</u>	2.65/3.99	2.57/3.75
	0.5	2.75/4.10	2.65/3.90	<u>2.68/3.98</u>	2.61/3.83
	Avg	2.56/3.76	2.52/3.69	<u>2.50/3.66</u>	2.44/3.54

The values reported in each cell denote MAE/RMSE. The 'Avg' row represents the average performance across different missing rates for each method. Bold and underlined values show the best and second-best values, respectively.

that of GAIN, it provides superior performance in handling incomplete multimodal data, making the tradeoff between accuracy and computational overhead well justified for practical applications. Unlike the multistep diffusion method like CSDI, our iMMAir controls the recovery effect through a simple one-step diffusion process. This simplification helps reduce the model's inference FLOPs from teraflops to gigaflops, which boosts computational efficiency and enhances practical usability in real-time applications. This demonstrates the versatility of our method in handling different modality types. Other modalities, such as land use data, can also be incorporated into our approach for air pollution prediction.

5.6.2 Validation of Spatial-Temporal Deep Learning Methods. To effectively validate the adaptability of our approach across different air quality datasets, we assess our model's performance against some spatiotemporal air quality prediction methods using the two public datasets. Table 3 shows our model's superior performance in handling incomplete air quality data across diverse missing rates (η). On average, our model surpasses the second-best baseline by approximately 5.6% and 2.8% on the two datasets, respectively. While these spatiotemporal deep learning methods such as Airformer perform well with complete datasets ($\eta = 0.0$), their performance markedly declines in severely missing cases, primarily due to the absence of well-performing modal recovery modules. Importantly, these spatiotemporal models can be integrated into our framework as the predictive model, with performance enhancements achieved by incorporating our recovery module. The results further verify the effectiveness of our iMMAir in incomplete multimodal air quality prediction, thereby enhancing its adaptability and reliability across diverse datasets.

6 Conclusion

In this work, we proposed a novel diffusion-driven incomplete multimodal learning approach (iMMAir) for air quality prediction, which has not been well explored in previous multimodal air quality studies. We designed a conditional diffusion-driven recovery approach to infer the distribution of missing data under two conditional constraints of temporal correlation and semantic consistency. To alleviate the modality imbalance problem, we employed an adaptive gradient modulation strategy to control the balanced optimization process of each modality. Experimental results demonstrated that our iMMAir achieves state-of-the-art performance across diverse missing data scenarios for regression and classification tasks. However, the applicability of our approach to complete multimodal air quality prediction tasks is still insufficient due to the data incompleteness issue of the dataset we collected. In our future work, we plan to collect a more expansive dataset across different data sources, spanning broader geographic regions and extended periods. In addition, we can incorporate some online learning techniques, which allow the model to continuously update and adapt to new incoming data without the need for full retraining. Moreover, we will develop a spatiotemporal large language model that leverages the inherent advantages of the large language model in processing large amounts of data and extracting complex patterns for air quality analysis. Then, we will further integrate physical knowledge of air quality to refine the spatial and temporal representations, thereby enhancing the model's interpretability and improving the predictive performance.

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